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Synthesis modification of hydroxyapatite surface for ethanol conversion: The role of the acidic/basic sites ratio

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ABSTRACT

Several studies have investigated the acidic/basic sites of the hydroxyapatite (HAP) surface. However, their specific role in the catalytic properties of HAP is still discussed. Here, we investigated different preparation methods (ultrasound, microwave, and autoclave) of HAP and how they affect the catalytic activity for ethanol conversion. The synthesis methods changed the number and distribution of acidic/basic sites, affecting the product selectivity. The best performance toward condensation products was observed for the microwave-treated catalyst. We also report a density functional theory (DFT) investigation on the reaction pathways of twelve steps of ethanol conversion on the HAP(0001) surface. Our DFT results suggest that the availability of basic sites can facilitate acetaldehyde formation. Furthermore, our theoretical insights indicate that the balance of acidic/basic sites contributes to the C–C bond formation, which is consistent with our experimental observations.

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1. Introduction

Hydroxyapatite (HAP, $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$) is a calcium hydroxyphosphate that accounts for approximately 70% of the mass of bones and teeth [1], making it a prominent material for biomedical applications [2]. Synthetic HAP has also been investigated as a catalyst in reactions of environmental interest, such as the oxidation of volatile organic compounds [3], the incineration of hazardous wastes [4], and the removal of heavy metals from contaminated soil and wastewater [5]. The cracking of organic amides [6] was one of the first HAP catalytic applications, but recently, this solid has been tested in various reactions, including pentose synthesis [7], levulinic acid hydrogenation [8], and ethanol upgrading [9–14].

The key feature for selecting HAP as a catalyst is the coexistence of acidic and basic sites in single crystals, allowing the co-adsorption of molecules with distinct acid-base properties [9,13]. Furthermore, the distribution of both sites on the HAP surfaces can be controlled by adjusting the bulk Ca/P ratio during precipitation synthesis, which involves wet reactions between Ca and P precursors with temperature and pH control [15]. For example, a Ca/P

ratio of 1.50 (low ratio) makes HAP an acidic catalyst, while an increasing in the Ca/P ratio to 1.67 (high ratio) makes HAP a basic catalyst. Thus, a fine control of the Ca/P ration between 1.50 and 1.67 can yield HAP catalysts with both acidic and basic properties [9].

The importance of the acidity and basicity of HAP is evident in the conversion of ethanol [9,16,11,17]. In this catalytic system, the conversion and product selectivity can be tuned by varying the surface distribution of acidic and basic sites. For example, ethylene is obtained with high selectivity in the conversion of ethanol over acidic HAP (low Ca/P ratio), while acetaldehyde is the main product obtained over basic HAP (high Ca/P ratio). Moreover, an even balance of acidic and basic sites with intermediate-weak adsorption strength results in the Guerbet coupling of ethanol into higher alcohols [18–20,13,21]. The Guerbet coupling involves a series of consecutive reactions: dehydrogenation, aldolization, dehydration, and hydrogenation [22,23]. From this reaction mechanism, C–C bonds are formed by the coupling of carbonyl compounds, thus leading to the production of C_4+ molecules.

Thus, the atomistic understanding of HAP surface properties is key to improving the design and activity of HAP catalysts for ethanol upgrading. Consistently, spectroscopy studies have suggested the involvement of Ca^{2+} ions as Lewis acidic sites, O atoms of PO_4^{3-} and OH^- groups as Lewis basic sites, and $\text{PO}-\text{H}$ as Brønsted

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acidic sites [24,11,25]. However, the specific role of HAP acid-base pairs in ethanol conversion remains in debate. In this regard, theoretical calculations can be useful to investigate how the HAP surface species affect the adsorption strength and the sequence of reaction steps.

At present, several studies based on density functional theory (DFT) calculations have been conducted to evaluate the HAP bulk and surface properties [26–31]. Regarding HAP adsorption sites, the exposed Ca^{2+} ions have been computed as the most HAP acidic sites from CO adsorption on different surface terminations [32,33]. Furthermore, Ca^{2+} ions were identified as energetically preferred sites for the adsorption of H_2O molecules, with basic O atoms in neighboring PO_4^{3-} groups assisting in the stabilization of the adsorbate via the reinforcement from H-bond interaction [34]. Recently, Ca^{2+} ions and PO_4^{3-} groups were also found as the energetically preferred Lewis acidic and Lewis basic adsorption sites, respectively, for the adsorption of various probe molecules [31].

Ultrasonic- and microwave-assisted syntheses have been widely used in the preparation of HAP [35]. Besides the low energy consumption compared to conventional heating techniques, microwave irradiation also offers a rapid bulk heating capability, which can decrease the particle size distribution of the solids [36–38]. Alternatively, the use of ultrasound after the precipitation process can disperse and deagglomerate the particles [39,40]. Although these techniques can change the morphology of HAP, it is unclear if they affect the Ca/P ratio by generating different surface terminations and, thus, change the catalytic activity of HAP. To the best of our knowledge, no theoretical study has been conducted on ethanol conversion over HAP catalysts. Therefore, the understanding of surface modifications in the synthesis of HAP for catalytic applications, as well as the effect of its acid-base pairs on the reactive steps of aldol condensation remains still unsatisfactory.

In the present study, we address the effect of different post-synthesis treatments on the distribution of HAP acidic and basic sites and its performance as a catalyst for ethanol conversion. Five HAP catalysts were prepared and tested in the gas-phase conversion of ethanol. The bulk composition was the same for all solids, but the distribution of acidic and basic sites on the HAP surface depended on the conditions of the post-synthesis treatments. An overview of the catalytic conversion of ethanol shows that the Guerbet reaction to form 1-butanol is favored for HAP catalysts with an even balance of acidic and basic sites, while the ethanol dehydration to ethylene is favored over more acidic HAP catalysts. Furthermore, DFT calculations were used to model the aldol condensation reaction pathway on the HAP(0001) surface. Finally, the intrinsic activation barriers were calculated considering the adsorption of reactants on different acidic and basic sites, allowing us to establish a sequence of reactions that is consistent with the experimental results.

2. Materials and methods

2.1. Synthesis of hydroxyapatite

The preparation of stoichiometric HAP is described elsewhere in detail [13]. In short, 50 mL of $(\text{NH}_4)_2\text{HPO}_4$ 0.6 M (Merck, 99%) was added dropwise to 50 mL of $\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$ 1.0 M (Merck, 99%) at 323 K under constant stirring. The pH of the mixture was controlled by the addition of NH_4OH (Merck, 25%) and kept at 10.5. The final slurry was kept in the mother liquor for 24 h before being vacuum filtered (no treatment, NT) or further treated post-synthesis. Four post-synthesis treatments, Table 1, were carried out, namely, ultrasound (US), autoclave (AC), microwave irradiation for 10 min (MW_{10}), and microwave irradiation for 30 min

Table 1

Post-synthesis treatments for the preparation of HAP catalysts.

Catalyst	Technique	Conditions
NT	No treatment	Room temperature/24 h
US	Ultrasound	Room temperature/2 h
AC	Autoclave	353 K/2 h
MW_{10}	Microwave	353 K/500 W/10min
MW_{30}	Microwave	353 K/500W/30min

(MW_{30}). After filtration and washing of the precipitate, HAP was dried at 393 K for 2 h. The samples were then calcined from room temperature to 873 K with a ramp of 1 K min^{-1} , and remained at 873 K for 5 h.

The crystalline structure of HAP was determined by powder X-ray diffraction (XRD) in a Philips PW 1710 diffractometer with $\text{CuK}\alpha$ radiation ($\lambda = 0.15406$ nm) at 40 kV and 30 mA. The diffraction patterns were taken from $2\theta = 20^\circ$ to 55° with a step of 0.01°s^{-1} . Calcium (Ca) and phosphorus (P) contents were determined by electron dispersive spectroscopy (EDS) and X-ray photoelectron spectroscopy (XPS). The bulk composition was determined after sample homogenization using a scanning electron microscope (SEM TESCAN, VEGA LMU) coupled to an energy dispersion spectrometer (EDS X-MAX 20, Oxford). XPS spectra were obtained using a VSW HA-100 spherical analyzer and non-monochromated $\text{AlK}\alpha$ radiation ($h\nu = 1486.6$ eV). The high-resolution spectra were measured using a constant analyzer pass energy of 44 eV with a full width at half-maximum (FWHM) of 1.6 eV for the Au $4f_{7/2}$ line. The pressure was kept below 2×10^{-8} mbar during the analysis, while the samples were fixed to a stainless-steel sample holder with double-faced conducting tape and analyzed without further preparation. Surface charging was corrected by shifting all spectra so that the C1s line due to adventitious carbon remained at 284.6 eV. Curve fitting was performed using Gaussian functions and the Shirley background [41] was subtracted from the data.

The thermal stability of the samples was evaluated by thermogravimetric analysis (TGA) in a SHIMADZU TGA-50M equipment with a microanalytical balance (Mettler Toledo MX5). The samples were heated from room temperature to 1073 K with a ramp of 10 K min^{-1} and an air flow of 50 mL min^{-1} . The textural properties were measured by N_2 physisorption at 77 K in a Micromeritics ASAP 2020 equipment. Before the analysis, samples (ca. 200 mg) were pretreated at 673 K for 24 h. The specific surface area was calculated by the BET method, while the mean volume and pore size were calculated by the BJH method. The density of acidic and basic sites was determined by NH_3 and CO_2 temperature-programmed desorption (TPD) in a Micromeritics AutoChem II 2920 equipment. Before the TPD analysis, the samples were pretreated at 673 K, saturated with probe molecule at 308 K (CO_2) or 323 K (NH_3), and heated up to 873 K at a rate of 10 K min^{-1} .

2.2. Ethanol conversion

The ethanol conversion on HAP catalysts was carried out through a single-step gas-phase reaction (at atmospheric pressure). A U-shaped quartz tube reactor with an inner diameter (ID) of ≈ 5 mm was used inside a vertical muffle furnace (ID ≈ 2 in), as described previously [13]. The reaction line was heated to its full extent, i.e., before (at 363 K) and after (383 K) the reactor. Reactions were carried out at 573 K, 598 K, 623 K, 648 K, and 673 K with a weight hourly space velocity (WHSV) for ethanol equal to 27.8 h^{-1} . The catalysts (80 mg) were pretreated inside the reactor at the reaction temperature for 1 h under N_2 flow (White Martins, 99.999%). Nitrogen was bubbled into absolute ethanol (Merck, 99.9%). The gaseous mixture flowed through a

saturator at the required temperature of 319.45 K for an ethanol concentration of 25% vol, Section S4. The reactor outflow gases were analyzed in a gas chromatograph (Agilent Technologies, 7890A) equipped with a flame ionization detector (FID). The column used was a DB1 (length = 60 m, ID = 0.53 mm, film = 1 μ m). For the identification and quantification of the ethanol reaction products, standard hydrocarbons (oxygenated and non-oxygenated), as well as response factors for flame ionization detectors [42–44] were used. The ethanol conversion, X (%), the product selectivity, S_i (%), and the turnover rate, TOR (h^{-1}), were calculated according to the following equations:

$$X = \left(1 - \frac{F_E^{\text{out}}}{F_E^{\text{in}}} \right) \times 100, \quad (1)$$

$$S_i = \left(1 - \frac{F_i^{\text{out}}}{F_{\text{total}}^{\text{out}}} \right) \times 100, \quad (2)$$

$$TOR = \frac{r}{[L]} = \frac{\xi}{W \times [L]}, \quad (3)$$

where F_E^{in} and F_E^{out} (mmol h^{-1}) are the input and exit molar flow rates of ethanol, F_i^{out} (mmol h^{-1}) is the exit molar flow rate of product i , $F_{\text{total}}^{\text{out}}$ (mmol h^{-1}) is the total exit molar flow rate, r ($\text{mmol h}^{-1}\text{g}^{-1}$) is the reaction rate, $[L]$ (mmol g^{-1}) is the concentration of active sites on the catalyst surface, estimated as the total number of acidic and basic sites that desorb the probe molecules (CO_2 or NH_3) up to 673 K in the TPD analysis (Table S2, SM), ξ (mmol h^{-1}) is the extent of reaction, and W (g) is the catalyst mass loaded into the reactor.

2.3. Theoretical calculations

All total energy calculations were based on spin-polarized DFT [45,46] adopting the semilocal Perdew–Burke–Ernzerhof (PBE) formulation for the exchange–correlation functional [47]. The semiempirical D3 van der Waals (vdW) correction proposed by Grimme [48] was used to improve the description of long-range vdW interactions. We employed the Vienna *ab initio* simulation package (VASP), version 5.4.1, which implements the all-electron projected augmented-wave (PAW) method [49–52]. The electron–ion interactions were described by the PAW projectors provided within VASP along with a plane-wave cutoff energy of 489 eV, which is 12.5% higher than the largest recommended value considering all chemical elements (C, Ca, H, O, and P).

The initial geometries of the isolated gas-phase molecules were obtained from the PubChem database [53] and optimized using a 20 Å cubic box. The clean hexagonal HAP(0001) surface was modeled using a repeated slab geometry with a thickness of four layers (one formula unit, $\text{Ca}_5(\text{PO}_4)_3(\text{OH})$, per layer and Ca/P ratio of 1.67), and a vacuum region of 15 Å. To search for the lowest energy configurations, 30 distinct structures were built for each adsorbed system using a 1×1 surface unit cell with $a_0 = b_0 = 9.497 \text{ \AA}$ (88 atoms within the unit cell). The adsorption structures were built by symmetrically placing adsorbates on both sides of the slab with respect to its center of inversion. As a result, dipole correction was no longer required. Once the lowest energy configurations were identified, the optimized adsorbed structures were placed on the surface 2×2 unit cell ($a_0 = b_0 = 18.995 \text{ \AA}$, 352 atoms within the unit cell) for a final structure optimization. All optimized structures were calculated allowing all atoms to relax through the conjugate gradient algorithm. The total energy convergence criterion was set to 10^{-5} eV with a force convergence of $0.025 \text{ eV \AA}^{-1}$ on each atom. Only the Γ -point was employed for the Brillouin zone integration.

The conversion of ethanol over stoichiometric HAP(0001) surface was modeled as a sequence of elementary steps. Different reaction pathways of 12 elementary steps were evaluated for the dehydrogenation and dehydration of ethanol, the formation of enolate, and the aldol condensation of acetaldehyde. The intrinsic activation energy of each elementary step was calculated using the adsorption energies of the adsorbates according to the unity bond index–quadratic exponential potential (UBI–QEP) approach developed by Shustorovich [54–56]. All adsorption and intrinsic activation energies presented in this study were calculated using a 2×2 surface unit cell to minimize the interaction between periodic images. The complete set of optimized structures, as well as the equations for calculating the intrinsic activation energies, are available in the Supplementary Material (SM), Section S6 and S7.

3. Results and discussion

3.1. Characterization of catalysts

All samples had the same diffraction patterns characteristic of hydroxyapatite (Figure S1, SM). HAP peaks were indexed according to a standard pattern (ICDD PDF 1-89-6440). No impurity phase was observed. The sharp and well-defined peaks suggest highly crystalline structures. The position of the 2θ angles for all solids deviated less than 0.9% from the reference crystal. All solids presented similar lattice parameters; $a_0 = b_0$ lattice constants ranged from 9.3734 Å to 9.4732 Å, and c_0 ranged from 6.8576 Å to 6.8978 Å, Table 2. The small variations observed are probably related to the relaxation of the unit cell due to the post-synthesis treatments. The ultrasound- and microwave-assisted synthesis considerably decreased the preparation time of the solids — from 24 h for the sample without treatment to 10 min for the sample exposed to microwave radiation. This result corroborates previous studies that point to faster nucleation and crystal growth in samples submitted to post-synthesis treatment [58–60,37].

The thermogravimetric profile of the HAP samples showed a single stage of mass loss (Figure S2, SM). The mass loss is associated with the removal of physically adsorbed water [61]. Type-IV (a) isotherms, characteristic of mesoporous materials ($2 \text{ nm} < D_p < 50 \text{ nm}$), were observed for all solids in this study (Figure S3, SM). However, post-synthesis treatments changed the morphology of the samples. The specific surface areas ranged from $32 \text{ m}^2\text{g}^{-1}$ to $41 \text{ m}^2\text{g}^{-1}$, a change of -13.5% to $+10.8\%$ when compared to the untreated solid and the average pore size increased with post-synthesis treatment.

The post-synthesis treatment did not affect the bulk composition of HAP samples. The Ca/P ratio measured in the bulk was determined based on the average values in the EDS and was equal to 1.66 for all solids. The elements present on the surface of the solids and their chemical state were identified by XPS. The XPS survey spectra of all samples were similar and identified calcium, phosphorus, and oxygen as major components (Figure S4, SM). A small signal of carbon was detected and associated with adventitious carbon due to adsorbed carbon impurities [13,62,63]. Fluorine was also detected as an impurity and associated with the PTFE magnetic stirring bar used in HAP synthesis. The electron binding energies of $\text{Ca}2p_{1/2}$ for all samples (349.9 eV to 350.7 eV) are commonly attributed to ionic Ca^{2+} , such as in CaO and CaCO_3 species [64]. The electron binding energies from 346.3 eV to 347.2 eV for $\text{Ca}2p_{3/2}$ were associated with Ca^{2+} ions in different calcium species, such as $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$ [65–68], $\text{Ca}(\text{HCOO})_2$ [69], CaO [64,66,69], CaCO_3 [64,69], $\text{Ca}_3(\text{PO}_4)_2$ [69] and CaHPO_4 [66,69]. The signal of $\text{P}2p$, registered between 132.6 eV to 133.2 eV, suggested the presence of PO_4^{3-} species, as expected in HAP [65,66,68] or $\text{Ca}(\text{H}_2\text{PO}_4)_2$ [69]. The doublet peak of oxygen (530.5

Table 2

Crystallographic and morphological properties of HAP catalysts. Lattice parameters, $a_0 = b_0$ and c_0 (Å); unit cell volume, V (Å³); particle size calculated by the Williamson–Hall method [57] with FWHM values obtained by Rietveld refinement, D (nm); specific surface area, S_{BET} (m²g^{−1}); pore size, D_p (nm); and pore volume, V_p (cm³).

Catalyst	$a_0 = b_0$	c_0	V	D	S_{BET}	D_p	V_p
NT	9.4047 ± 0.0078	6.8881 ± 0.0063	527.6 ± 1.0	20.823 ± 0.462	36	27.5	0.30
US	9.4210 ± 0.0185	6.8792 ± 0.0149	528.8 ± 2.3	18.937 ± 0.427	31	37.6	0.31
AC	9.4732 ± 0.0232	6.8978 ± 0.0187	532.0 ± 2.9	20.176 ± 0.423	34	35.1	0.33
MW ₁₀	9.3734 ± 0.0053	6.8615 ± 0.0043	522.1 ± 0.6	19.809 ± 0.478	41	38.5	0.41
MW ₃₀	9.3897 ± 0.0165	6.8576 ± 0.0133	523.6 ± 2.0	19.710 ± 0.398	35	40.4	0.44

eV to 533.0 eV) was assigned to O^{2−} species and matches the most likely compounds suggested by the calcium signal. The electron binding energies of P2p, Ca2p and O1s were used to estimate the surface chemical composition. As observed in Table 3, the surface Ca/P atomic ratio is lower than the bulk Ca/P ratio, indicating that the surface is calcium-deficient at the outermost layers. This behavior was previously observed in several studies and seems to be a particular feature of hydroxyapatite [70,62,9,71]. Even though the bulk composition was the same for all solids, the surface chemical composition depended on post-synthesis treatment.

The desorption profiles of CO₂ and NH₃ for all solids had three peaks between 300 K and 800 K (Figure S6, SM). In this work, we classified as weak sites those that desorb up to 473 K in the TPD analysis, intermediate sites as those that desorb between 473 K and 673 K, and strong sites as those that desorb above 673 K. The total number of acidic sites increased up to 26.4% when compared to the untreated solid, except for the solid MW₁₀, which had the same total surface acidity as the untreated solid. The distribution of strongly adsorbing acidic sites is similar for all solids, with approximately half of the total number attributed to weak acidic sites. The surface basicity also changed with the post-synthesis treatments – the change in the number of basic sites ranged between −8.0% and +30.7% when compared to the untreated sample, Table 3. The differences in acidity and basicity may be a result of the variation of the Ca/P ratio on the surface. The ratio of the number of basic sites by the number of acid sites was assessed using two criteria, namely, the ratio between the total number of basic and acidic sites (B/A) and the ratio between intermediate-weak adsorption basic and acidic sites (B/A), i.e. sites where CO₂ or NH₃ was desorbed up to 673 K in the TPD analysis. If only total sites were considered, then all solids would be basic catalysts, as expected for a solid with Ca/P of 1.66. However, when only intermediate-weak adsorption sites are taken into account, HAP can be either a basic or an acidic catalyst. This behavior is discussed later in the catalytic tests with ethanol.

3.2. Catalytic conversion of ethanol

The catalytic activity of HAP is influenced by the distribution and adsorption strength of its acidic and basic sites and, therefore, by post-synthesis treatment. To ensure the kinetic control of catalytic rate measurements [72], the reactor was operated in conversions below 20% from 573 K to 673 K (Table 4). A possible mass transfer limitation was assessed using the Weisz–Prater criterion (Section S4, SM). For all evaluated conditions, the Weisz–Prater criterion was several orders of magnitude below unity, strongly suggesting no internal or external mass transfer limitation, even at higher conversions. Blank reactions were conducted under the same conditions as the reactions reported in this study (Table S2, SM). Acetaldehyde formation derived from the autothermic decomposition of ethanol was detected only in small amounts (up to 0.54% at 673 K), thus not affecting the conclusions drawn in this study.

The ethanol conversion was favored as the B/A ratio increased from 0.74 to 1.04. Interestingly, the most basic catalyst, AC with

$B/A = 1.21$, was not the most active for ethanol conversion. The ethanol conversion and product selectivity varied significantly with the catalyst. The main products resulting from ethanol conversion over HAP catalysts were acetaldehyde and 1-butanol, Fig. 1. Overall, the combined selectivity for these two products varied from 63% to 73%. The other products observed were ethylene, 1,3-butadiene, ethyl ether, crotonaldehyde, 2-buten-1-ol, butanal and 1-hexanol. Non-identified heavier products were also detected with selectivity lower than 10%, corresponding up to 8.1% in the overall carbon balance (Table S4, SM). No catalyst deactivation was observed within 4 h of time-on-stream. By increasing the temperature, the selectivity toward 1-butanol reaches a maximum and decreases as 1-butanol undergoes further condensation reactions to produce 1-hexanol and 2-ethyl-butanol.

No direct correlation was observed between the total number of acidic and basic sites and product selectivity. However, by considering only weak and medium sites, a trend could be observed. The ethylene formation was favored by increasing the number of acidic sites on the surface of HAP. Thus, at temperatures equal or higher than 598 K, the maximum selectivity to ethylene was observed at the lowest B/A ratio, i.e., for the US catalyst. The selectivity toward acetaldehyde is strictly linked to further condensation pathways. Even if a trend between acetaldehyde selectivity and basic sites is expected, the presence of both acidic and basic sites is required for the Guerbet reaction. The lower selectivity toward acetaldehyde and the increase in the amounts of crotonaldehyde, 2-buten-1-ol and 1-butanol are consistent with previous observations for 1-butanol production via aldol condensation of two aldehydes [9,12]. These trends are clearly observed for the MW₁₀ catalyst ($B/A = 1.00$) – as the temperature increased, the selectivity toward 1-butanol increased while toward acetaldehyde decreased.

Even though Fig. 1 shows the results of the selectivity for each reaction temperature, the ethanol conversion for each catalyst is different at a given temperature (Table 4). At lower temperatures (e.g., 573 K and 593 K), the difference in the conversion values are greater than at higher temperatures, which could lead to significant fluctuations. A better way to compare the selectivity profile is using the same conversion. Therefore, we used the experimental values of the turnover rate to estimate the product composition and their selectivity at a determined ethanol conversion (Figure S9, SM). Lower B/A values, measured for the catalysts US and NT, led to the lowest selectivities toward 1-butanol whereas the catalysts MW₁₀, MW₃₀, and AC are the most selective. The basic sites seem to be more relevant and a balance of acidic and basic sites is necessary for the Guerbet condensation.

The proposed reaction scheme for ethanol conversion is presented in Fig. 2. We acknowledge the possibility of 1-butanol formation via isomerization of 2-buten-1-ol to butanal followed by hydrogenation to 1-butanol. However, at lower temperatures, 573 K and 598 K, despite the reactions showing considerable selectivity toward 1-butanol, butanal was not detected in significant amounts. Furthermore, the inclusion of two additional reaction steps (2-buten-1-ol → butanal → butanol) would result in an increased number of adjustable parameters and likely instability of the fit.

Table 3

Surface composition and surface distribution of acidic and basic sites on HAP catalysts. Surface molar ratio of calcium and phosphorous, Ca/P; total number of basic sites, B ($\mu\text{mol}_{\text{CO}_2}\text{m}^{-2}$); total number of acidic sites, A ($\mu\text{mol}_{\text{NH}_3}\text{m}^{-2}$); ratio between the total number of basic and acidic sites, B/A ; ratio between the weak-intermediate basic and acidic sites, B/A .

Catalyst	Ca/P	B	A	B/A	B/A
US	1.47	17.75	8.92	1.99	0.74
NT	1.37	16.24	7.71	2.11	0.94
MW ₁₀	1.38	13.32	7.46	1.79	1.00
MW ₃₀	1.34	16.10	8.41	1.91	1.04
AC	1.45	21.23	9.75	2.18	1.21

Table 4

Ethanol conversion, X (%), on HAP catalysts at different reaction temperatures.

	573 K	598 K	623 K	648 K	673 K
US	1.07	2.43	4.09	7.44	13.35
NT	0.94	2.72	4.29	9.06	17.23
MW ₁₀	2.80	4.54	6.41	9.54	17.08
MW ₃₀	5.40	5.64	9.70	11.72	17.84
AC	3.29	5.56	7.30	11.76	17.61

A better way to compare the performance of the catalysts is by the turnover rate, Table 5. Additional details regarding the expressions for calculating the extent of reaction are provided in the SM, Section S4. The catalyst with the higher relative acidity (US) had the highest TOR values for the dehydration reactions, i.e., ethylene and 1,3-butadiene formation. At lower temperatures, from 573 K to 623 K, the TOR values of the reactions toward butanol production, i.e., reactions 3 to 6 in Fig. 2, increased for the catalysts prepared with some post-synthesis treatment. The MW₃₀ catalyst was significantly more active for ethanol dehydrogenation (r_3) and butanol formation (r_6) than any other catalysts tested from 573 K to 648 K. The TOR of acetaldehyde formation over MW₃₀ was 5 times higher than that observed over NT catalyst at 573 K. Also, the TOR for butanol formation over MW₃₀ was ≈ 2 times higher than that over NT catalyst at 598 K. At 673 K, the NT catalyst appears to be the most active catalyst for acetaldehyde production. However, the solid MW₃₀ is the only catalyst to have a significantly higher TOR , mainly toward the production of butanal, 1-hexanol and heavier products at this temperature. Catalysts treated with post-synthesis treatments are more active in ethanol dehydrogenation and butanol formation compared to NT catalyst from 573 K to 623 K. These observations suggest that post-synthesis treatment plays a beneficial role in catalysis during the coupling of ethanol.

According to the reaction scheme proposed in Fig. 2, the ethanol molecule participates in four reactions: dehydration to ethylene or diethyl ether, dehydrogenation to acetaldehyde, and reaction with 1-hexanol. The ratio between the TOR of acetaldehyde formation and all ethanol conversion reactions, i.e., $r_3/(r_1 + r_2 + r_3 + r_{11})$, has a very small variation even for significantly different conversions — equal to 0.85 ± 0.03 at 573 K, 0.82 ± 0.03 at 598 K, 0.84 ± 0.02 at 623 K, 0.82 ± 0.04 at 648 K, and 0.78 ± 0.03 at 673 K. This shows that the main reaction of ethanol conversion is the dehydrogenation to acetaldehyde and this ratio did not change within the temperature range studied.

Similarly, there are four reactions involving the conversion of acetaldehyde, i.e., r_4, r_8, r_9 and r_{10} . The ratios $r_4/(r_4 + r_8 + r_9 + r_{10})$ and $r_9/(r_4 + r_8 + r_9 + r_{10})$ range from 0.69 ± 0.06 and 0.28 ± 0.07 at 573 K, and from 0.56 ± 0.06 and 0.43 ± 0.06 at 673 K, suggesting that the reaction to produce 1-hexanol becomes more relevant as the temperature increases. It also suggests that the formation of 2-ethyl-1-butanol is irrelevant.

3.3. DFT-based reaction pathways

As previously stated, the HAP acidic and basic properties are strongly linked to its surface-exposed species. HAP(0001), reported as the thermodynamically preferred surface morphology [73], exhibits two types of adsorption sites: positively charged Ca^{2+} ions and negatively charged PO_4^{3-} groups, Fig. 3. A recent study evaluates the orientation of the OH^- groups along the HAP(0001) slab model [31], showing that after structural optimization, the OH^- groups become less exposed to the surface compared to Ca^{2+} and PO_4^{3-} groups. They do, however, play an important role in bulk and surface properties. Our catalyst model is limited to the choice of a single surface termination, HAP(0001), hence, it has limitations to describe experimental effects, such as defects, polycrystalline surfaces, and the extreme limits of the B/A ratio. However, using a single model with both Lewis acidic and Lewis basic regions to represent the experimental conditions could be advantageous. The presence of these two types of sites allowed for molecular-level insights into the impact of HAP adsorption sites on key ethanol conversion reaction mechanisms.

For the computational study of the HAP catalyst, the conversion of ethanol was modeled according to the sequence of elementary steps (ES) proposed in Table 6. More specifically, the ES 1a to 8 were used to understand the first steps of ethanol dehydration and dehydrogenation and discuss possible routes for acetaldehyde and ethylene formation. Meanwhile, ES 9a to 10b studied the deprotonation of acetaldehyde to form the enolate reaction intermediate, and ES 11a to 12 cover the C–C bond formation via an aldol condensation.

To evaluate the mechanistic differences that could be caused by variations in the surface distribution of acidic/basic sites, the reactants were adsorbed on Lewis acidic sites ($^*_{\text{A}}$; Ca^{2+} ions) and Lewis basic sites ($^*_{\text{B}}$; O atoms in PO_4^{3-} groups) exposed on the HAP(0001). Therefore, an acidic HAP catalyst with a lower B/A ratio was modeled by selecting optimized structures with the reactant adsorbed on the Lewis acidic site. Whereas, a basic HAP catalyst with a higher B/A ratio was modeled by selecting optimized structures with the reactant adsorbed on the Lewis basic site.

For all dissociation and combination reactions, the phenomenological UBI-QEP method was used to calculate the intrinsic activation energy, E_a . The magnitude of the intrinsic activation energy can be calculated using only the adsorption energy of adsorbates and the gas-phase bond dissociation energy as

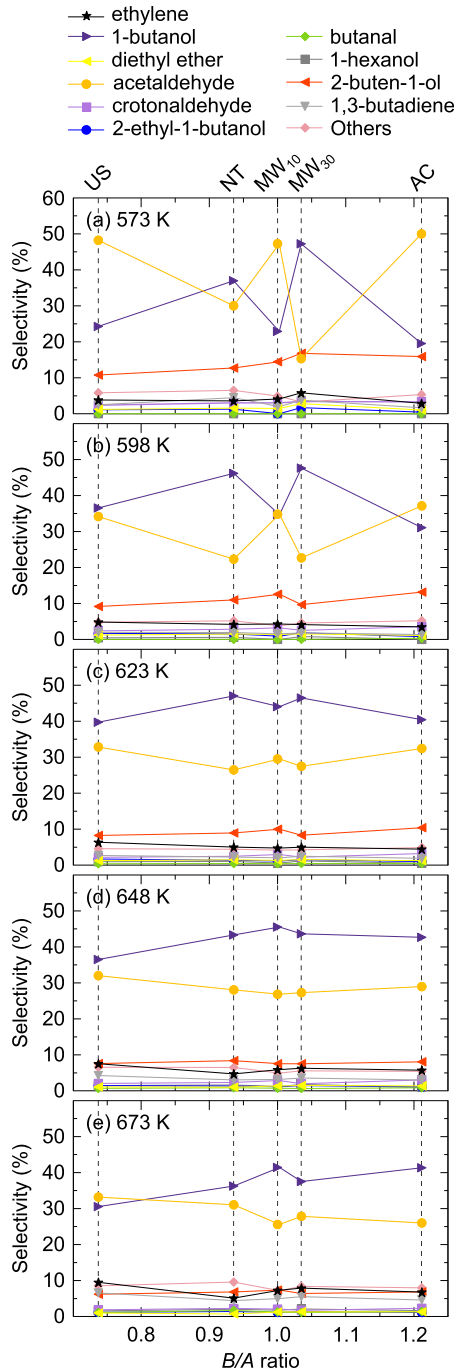


Fig. 1. Effect of surface site distribution on product selectivity. The reactions were conducted at (a) 573 K, (b) 598 K, (c) 623 K, (d) 648 K, and (e) 673 K. For the B/A ratio, only weak-intermediate basic and acidic sites were considered.

$$E_a = \frac{1}{2} \left(\Delta H_r + \frac{Q_a Q_b}{Q_a + Q_b} \right), \quad (4)$$

where

$$\Delta H_r = Q_{ab} + D_{ab} - Q_a - Q_b. \quad (5)$$

The surface reaction enthalpy, ΔH_r , corresponds to the sum of the enthalpies of the thermodynamic cycle: desorption of adsorbate species ab , Q_{ab} ; gas-phase bond dissociation, D_{ab} ; and adsorption of molecular fragments a and b , Q_a and Q_b . It should be stressed that the reactants and products are labeled in such a way that D_{ab} is always positive [56]. Then, for the dissociation reactions (ES 1a to

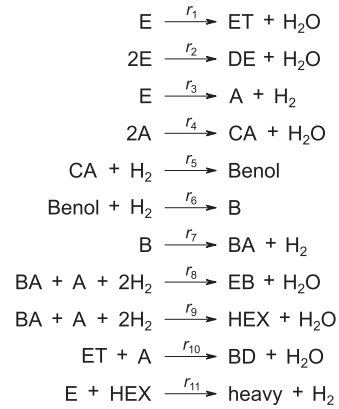


Fig. 2. Proposed reaction scheme for the conversion of ethanol on HAP catalysts. E = ethanol; ET = ethylene, DE = diethyl ether, A = acetaldehyde, CA = crotonaldehyde, Benol = 2-buten-1-ol; B = 1-butanol; BA = butanal; EB = 2-ethyl-1-butanol; HEX = 1-hexanol; BD = 1,3-butadiene, heavy = heavy products.

10b), the adsorbate species are labeled in the direct order of appearance, e.g., for the ES 1a : $ab = \text{CH}_3\text{CH}_2\text{OH}$, $a = \text{CH}_3\text{CH}_2\text{O}$, and $b = \text{H}$. Differently, for the combination reactions (ES 11a to 12), the adsorbate species are labeled in the reverse order of appearance, e.g., for the ES 11a : $ab = \text{C}_4\text{H}_7\text{O}_2$, $a = \text{CH}_2\text{CHO}$, and $b = \text{CH}_3\text{CHO}$.

The gas-phase bond dissociation energies, D_{ab} , were calculated as

$$D_{ab} = E_{\text{tot}}^a + E_{\text{tot}}^b - E_{\text{tot}}^{ab}, \quad (6)$$

where E_{tot}^a and E_{tot}^b are the total energies of the gas-phase molecular fragments a and b after the bond dissociation, and E_{tot}^{ab} is the total energy of the gas-phase molecule ab before the bond dissociation. To propose a consistent sequence of elementary steps from an energetic standpoint, the adsorption energies Q_a and Q_b were calculated considering the co-adsorption effect:

$$Q_a = \frac{1}{2} \left(E_{\text{tot}}^{a+b/\text{Sub}} - E_{\text{tot}}^{b/\text{Sub}} - 2E_{\text{tot}}^a \right), \quad (7)$$

$$Q_b = \frac{1}{2} \left(E_{\text{tot}}^{a+b/\text{Sub}} - E_{\text{tot}}^{a/\text{Sub}} - 2E_{\text{tot}}^b \right), \quad (8)$$

where $E_{\text{tot}}^{a+b/\text{Sub}}$ is the total energy of the coadsorbed system, while $E_{\text{tot}}^{a/\text{Sub}}$ and $E_{\text{tot}}^{b/\text{Sub}}$ are the total energies of the adsorbed fragments a and b at the zero coverage limit. The adsorption energy Q_{ab} was calculated as

$$Q_{ab} = \frac{1}{2} \left(E_{\text{tot}}^{ab/\text{Sub}} - E_{\text{tot}}^{\text{Sub}} - 2E_{\text{tot}}^{ab} \right), \quad (9)$$

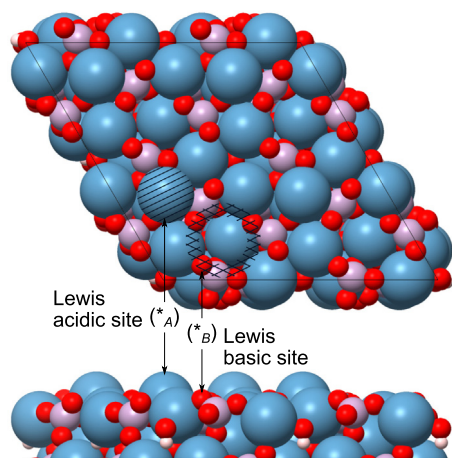
where $E_{\text{tot}}^{ab/\text{Sub}}$ is the total energy of the adsorbed ab , and $E_{\text{tot}}^{\text{Sub}}$ is the total energy of the clean HAP(0001) substrate. The terms $\frac{1}{2}$ and 2 in Equation 7 to 9 are consequences of adsorbing molecules on both sides of the slab. All values of adsorption and bond dissociation energies reported in this study were obtained at a temperature of 0 K without including vibrational and thermal corrections, which will be covered in future studies.

3.3.1. Dehydrogenation and dehydration of ethanol

The ethanol molecule was more strongly adsorbed on acidic sites than on basic sites exposed at the HAP(0001) surface ($-142.18 \text{ kJ mol}^{-1}$ vs $-62.63 \text{ kJ mol}^{-1}$, respectively). As the interaction between the hydroxy group of ethanol and the Lewis acidic Ca^{2+} ion increases, so does the Coulomb interaction, and hence, the adsorbed system increases its stability, Fig. 4. The stabilizing effect is also reflected by the intrinsic activation energy. All elementary

Table 5Temperature effect on the turnover rate, TOR (h^{-1}), of ethanol conversion reactions on HAP catalysts, see Fig. 2.

		r_1	r_2	r_3	r_4	r_5	r_6	r_7	r_8	r_9	r_{10}	r_{11}
573 K	US	0.22	0.04	13.48	4.34	4.25	3.87	3.02	0.04	2.98	0.09	2.98
	NT	0.24	0.05	10.29	3.61	3.52	3.14	2.04	0.04	2.01	0.13	2.01
	MW ₁₀	0.15	0.04	26.70	8.83	8.75	8.40	7.85	0.00	7.85	0.05	7.85
	MW ₃₀	0.16	0.05	52.30	17.70	17.64	17.37	16.60	0.03	16.57	0.06	16.57
	AC	0.14	0.03	26.03	8.55	8.45	7.97	7.39	0.01	7.37	0.05	7.37
598 K	US	0.67	0.12	31.31	10.83	10.60	9.69	6.08	0.17	5.91	0.19	5.87
	NT	0.48	0.12	29.93	10.92	10.70	9.83	6.18	0.12	6.06	0.15	6.01
	MW ₁₀	0.45	0.11	44.46	15.18	14.93	13.97	11.32	0.07	11.25	0.12	11.25
	MW ₃₀	0.45	0.12	56.47	19.72	19.53	18.80	15.19	0.14	15.05	0.14	14.99
	AC	0.40	0.09	44.98	15.25	14.97	13.89	11.36	0.06	11.27	0.11	11.27
623 K	US	2.08	0.28	54.41	19.28	18.79	16.89	7.74	0.39	7.27	0.62	7.06
	NT	1.40	0.28	49.40	18.41	17.94	16.21	7.11	0.26	6.76	0.43	6.57
	MW ₁₀	1.35	0.26	65.88	23.69	23.10	21.10	12.25	0.23	11.94	0.41	11.83
	MW ₃₀	1.29	0.27	98.26	34.34	33.96	32.52	24.47	0.26	24.14	0.43	23.93
	AC	1.19	0.23	61.71	21.83	21.23	19.28	11.69	0.18	11.43	0.35	11.33
648 K	US	6.22	0.56	100.23	35.27	34.19	30.18	10.86	0.76	9.74	2.24	8.98
	NT	3.95	0.52	107.16	39.78	38.60	34.31	12.12	0.67	11.01	1.54	10.18
	MW ₁₀	3.81	0.56	100.99	37.40	36.20	33.03	13.83	0.48	13.07	1.33	12.57
	MW ₃₀	3.73	0.56	122.80	43.83	43.09	40.24	23.66	0.55	22.87	1.34	22.20
	AC	3.27	0.51	100.78	36.43	35.30	32.27	16.22	0.38	15.53	1.12	15.14
673 K	US	15.31	0.92	174.32	58.09	56.32	50.42	21.26	1.10	19.11	6.25	17.63
	NT	10.01	0.71	202.69	71.60	69.24	62.07	23.88	1.45	20.68	4.65	18.58
	MW ₁₀	9.63	0.98	178.55	65.15	63.53	57.77	25.20	0.87	23.38	3.91	22.00
	MW ₃₀	9.59	0.94	187.10	65.56	64.34	59.77	32.88	1.01	31.07	3.94	29.59
	AC	8.26	0.93	153.32	56.24	54.59	49.63	19.57	0.80	17.81	3.33	16.61

**Fig. 3.** Investigated acidic and basic adsorption sites on HAP(0001). Lewis acidic and Lewis basic sites are indicated by simple hatch and crosshatched sections, respectively. Blue, pink, red, and white spheres represent Ca, P, O, and H atoms, respectively.

steps in which ethanol was initially adsorbed on acidic sites (ES 1a, 2a, 3a, and 4a) showed higher activation energies than those for ethanol adsorbed on basic sites (ES 1b, 2b, 3b, and 4b).

For the first deprotonation of ethanol, three possible bond cleavages were investigated: the O-H bond (ES 1a and 1b); the C-H bond in the α -carbon (ES 2a and 2b); and the C-H bond in the β -carbon (ES 4a and 4b). When ethanol was adsorbed on acidic sites, the activation energies were $132.95 \text{ kJ mol}^{-1}$, $158.74 \text{ kJ mol}^{-1}$ and $161.91 \text{ kJ mol}^{-1}$, respectively, suggesting that the cleavage of the O-H bond on the HAP(0001) surface is less energetic than the cleavage of C-H bonds. A similar trend was also observed for ethanol adsorbed on basic sites with intrinsic activation energies of $93.17 \text{ kJ mol}^{-1}$, $118.96 \text{ kJ mol}^{-1}$ and $122.13 \text{ kJ mol}^{-1}$, respectively.

On average, the ethanol O-H bond was the most affected upon adsorption on the HAP surface. For example, the interaction with

acidic sites resulted in the stretching of the O-H bond by 4.05% versus a contraction of the C-H bond in the α -carbon by -0.15% , while there is a tiny deformation of the C-H bond in the β -carbon by 0.07%. The average bond variations for ethanol on basic sites were 1.30%, 0.03% and 0.02%, respectively, indicating that a weaker interaction with the substrate has a slight impact on the adsorbate structure. In the gas-phase, the O-H bond dissociation is more energetic compared to the dissociation of the C-H bond in the α -carbon ($443.87 \text{ kJ mol}^{-1}$ vs $402.11 \text{ kJ mol}^{-1}$, respectively). However, the deprotonation of ethanol via cleavage of the O-H bond (ES 1a and 1b) is less energetic in the presence of the catalyst surface, which could be related to the observed stretching of the O-H bond distance.

Deprotonation via cleavage of the O-H bond resulted in molecular fragments with more negative values of co-adsorption energies. However, the subsequent deprotonation to form acetaldehyde after cleavage of the O-H bond (ES 5) requires less energy compared to the deprotonation after cleavage of the C-H bond in the α -carbon (ES 6). The intrinsic activation energies of these reactions were $13.05 \text{ kJ mol}^{-1}$ and $40.28 \text{ kJ mol}^{-1}$, respectively. Although both deprotonation routes were found to be exothermic, the surface reaction enthalpy calculated for the ES 5 was much more negative than the one calculated for ES 6 ($-61.47 \text{ kJ mol}^{-1}$ vs $-7.02 \text{ kJ mol}^{-1}$).

For the dehydration of ethanol to form ethylene, two possible reaction routes were evaluated: the cleavage of the ethanol C-OH bond (ES 3a and 3b) followed by the deprotonation of the $\text{CH}_3\text{-CH}_2$ reaction intermediate (ES 7); and the deprotonation of the ethanol β -carbon (ES 4a and 4b) followed by the cleavage of the C-OH bond of the $\text{CH}_2\text{CH}_2\text{OH}$ reaction intermediate (ES 8). For ethanol adsorbed on acidic and basic sites, the C-OH bond was stretched by 1.33% and contracted by -0.32% , respectively. The cleavage of the ethanol C-OH bond produces a very unstable reaction intermediate that is readily converted to ethylene requiring an intrinsic activation energy of 6.45 kJ mol^{-1} via an exothermic reaction at $-42.40 \text{ kJ mol}^{-1}$. In contrast, after the aforementioned deprotonation of the ethanol β -carbon, the subsequent cleavage of the C-OH bond in the ES 8 is an endothermic reaction at 50.13

Table 6

Proposed sequence of elementary steps on HAP(0001) for C–C bond formation via aldol condensation. Gas-phase bond dissociation energy, D_{ab} ; adsorption energy, Q_{ab} ; co-adsorption energies of molecular fragments, Q_a and Q_b ; surface reaction enthalpy, ΔH_f ; and intrinsic activation energy, E_a . All numbers are in kJ mol^{-1} .

	Elementary Steps (ES)	D_{ab}	Q_{ab}	Q_a	Q_b	ΔH_f	E_a
1a	$\text{CH}_3\text{CH}_2\text{OH}_A^* + \text{B}^* \rightarrow \text{CH}_3\text{CH}_2\text{O}_A^* + \text{H}_B^*$	443.87	−142.18	−491.60	−426.62	37.49	132.95
1b	$\text{CH}_3\text{CH}_2\text{OH}_B^* + \text{A}^* \rightarrow \text{CH}_3\text{CH}_2\text{O}_B^* + \text{H}_A^*$	443.87	−62.63	−491.60	−426.62	−42.07	93.17
2a	$\text{CH}_3\text{CH}_2\text{OH}_A^* + \text{B}^* \rightarrow \text{CH}_3\text{CHOH}_A^* + \text{H}_B^*$	402.11	−142.18	−294.96	−217.29	192.37	158.74
2b	$\text{CH}_3\text{CH}_2\text{OH}_B^* + \text{A}^* \rightarrow \text{CH}_3\text{CHOH}_B^* + \text{H}_A^*$	402.11	−62.63	−294.96	−217.29	112.81	118.96
3a	$\text{CH}_3\text{CH}_2\text{OH}_A^* + \text{B}^* \rightarrow \text{CH}_3\text{CH}_2^* + \text{OH}_A^*$	420.85	−142.18	−357.97	−557.95	46.16	131.36
3b	$\text{CH}_3\text{CH}_2\text{OH}_B^* + \text{A}^* \rightarrow \text{CH}_3\text{CH}_2^* + \text{OH}_B^*$	420.85	−62.63	−357.97	−557.95	−33.40	91.59
4a	$\text{CH}_3\text{CH}_2\text{OH}_A^* + \text{B}^* \rightarrow \text{CH}_2\text{CH}_2\text{OH}_A^* + \text{H}_B^*$	440.21	−142.18	−353.62	−266.61	171.81	161.91
4b	$\text{CH}_3\text{CH}_2\text{OH}_B^* + \text{A}^* \rightarrow \text{CH}_2\text{CH}_2\text{OH}_B^* + \text{H}_A^*$	440.21	−62.63	−353.62	−266.61	92.95	122.13
5	$\text{CH}_3\text{CH}_2\text{O}_A^* + \text{B}^* \rightarrow \text{CH}_3\text{CHO}_A^* + \text{H}_B^*$	88.26	−121.95	−214.71	−147.90	−61.47	13.05
6	$\text{CH}_3\text{CH}_2\text{O}_B^* + \text{A}^* \rightarrow \text{CH}_3\text{CHO}_B^* + \text{H}_A^*$	130.02	−134.64	−214.71	−147.90	−7.02	40.28
7	$\text{CH}_3\text{CH}_2^* + \text{A}^* \rightarrow \text{C}_2\text{H}_4^* + \text{H}_A^*$	169.47	41.07	−113.84	−107.58	−42.40	6.45
8	$\text{CH}_2\text{CH}_2\text{OH}_A^* + \text{B}^* \rightarrow \text{C}_2\text{H}_4^* + \text{OH}_A^*$	150.10	−143.98	−81.04	−224.41	50.13	54.83
9a	$\text{CH}_3\text{CHO}_A^* + \text{B}^* \rightarrow \text{CH}_2\text{CHO}_A^* + \text{H}_B^*$	408.52	−127.38	−452.81	−382.46	26.11	116.73
9b	$\text{CH}_3\text{CHO}_B^* + \text{A}^* \rightarrow \text{CH}_2\text{CHO}_B^* + \text{H}_A^*$	408.52	−35.99	−452.81	−382.46	−65.27	71.03
10a	$\text{CH}_3\text{CHO}_A^* + \text{B}^* \rightarrow \text{CH}_3\text{CO}_A^* + \text{H}_B^*$	380.41	−127.38	−311.81	−264.41	139.01	141.05
10b	$\text{CH}_3\text{CHO}_B^* + \text{A}^* \rightarrow \text{CH}_3\text{CO}_B^* + \text{H}_A^*$	380.41	−35.99	−311.81	−264.41	47.63	95.35
11a	$\text{CH}_3\text{CHO}_A^* + \text{CH}_2\text{CHO}_A^* \rightarrow \text{C}_4\text{H}_7\text{O}_A^{*+}$	6.43	−180.00	−111.04	−109.57	51.79	53.47
11b	$\text{CH}_3\text{CHO}_B^* + \text{CH}_2\text{CHO}_B^* \rightarrow \text{C}_4\text{H}_7\text{O}_B^{*+}$	6.43	−180.00	−142.66	−49.19	−5.65	15.47
12	$\text{C}_4\text{H}_7\text{O}_A^{*+} + \text{H}_B^* \rightarrow \text{C}_4\text{H}_8\text{O}_A^{*+}$	444.85	−203.47	−382.74	−505.77	−85.58	66.14

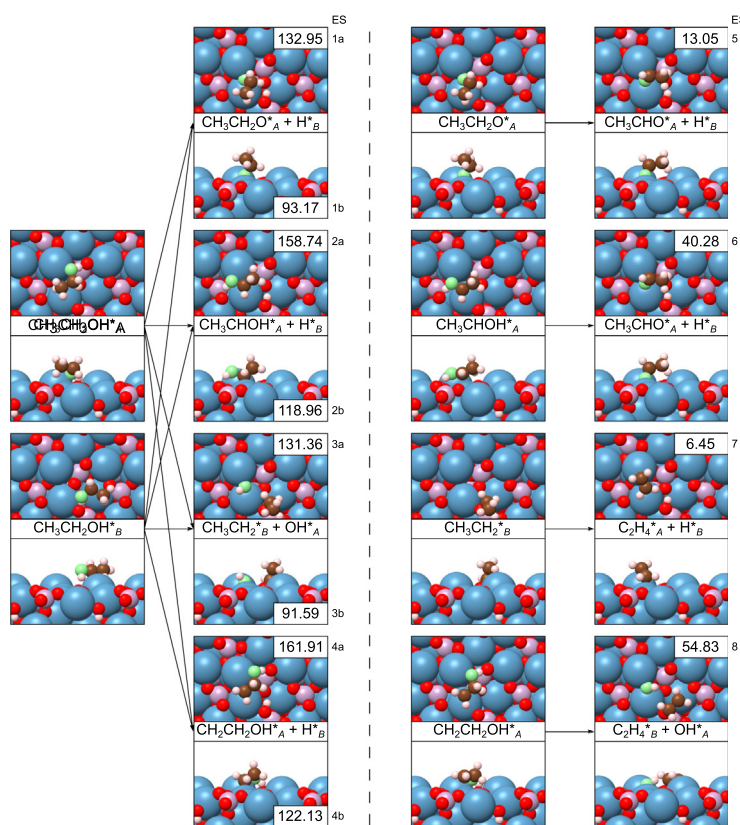


Fig. 4. Reaction pathways for dehydrogenation and dehydration of ethanol on HAP(0001). Numbers inside the white rectangles represent the intrinsic activation energy (kJ mol^{-1}) for each of the ES listed in Table 6. Blue, pink, and red spheres represent Ca, P, and O atoms in the substrate. Brown, light green, and white spheres represent C, O, and H atoms in the adsorbate. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

kJ mol^{-1} with a much higher intrinsic activation energy of $54.83 \text{ kJ mol}^{-1}$.

An overview of the ethanol reaction pathway, Fig. 4, indicates that the adsorption of ethanol on the HAP(0001) basic sites facilitates both the dehydrogenation to acetaldehyde and the dehydration to ethylene, with a significant decrease in intrinsic activation energies compared to the adsorption on acidic sites. Furthermore, the production of acetaldehyde may proceed via the cleavage of

the O–H bond in the hydroxy group, while ethylene may be obtained via the cleavage of the C–OH bond.

3.3.2. Enolate formation

Adsorption of acetaldehyde and its deprotonation to form the enolate were also evaluated on HAP(0001). The most stable adsorption configuration for the acetaldehyde molecule was found to be on the Lewis acidic Ca^{2+} ion via interaction with the O atom of

the carbonyl group, Fig. 5. In this configuration, the acetaldehyde adsorption energy ($\text{CH}_3\text{CHO}^*_\text{A}$, $-127.38 \text{ kJ mol}^{-1}$) was much more negative than on an adjacent basic site after surface diffusion ($\text{CH}_3\text{CHO}^*_\text{B}$, $-35.99 \text{ kJ mol}^{-1}$).

Then, the adsorbed acetaldehyde can be deprotonated at the α -carbon (ES 9a and 9b) to form enolate or at the carbonyl group (ES 10a and 10b). For the deprotonation at the α -carbon, the reaction is endothermic ($\Delta H_r = 26.11 \text{ kJ mol}^{-1}$) with an intrinsic activation energy of $116.73 \text{ kJ mol}^{-1}$ when acetaldehyde is adsorbed on acidic sites, but exothermic ($\Delta H_r = -65.27 \text{ kJ mol}^{-1}$) with an intrinsic activation energy of $71.03 \text{ kJ mol}^{-1}$ when acetaldehyde is adsorbed on basic sites. For the deprotonation of the carbonyl group, the reaction is endothermic for both acidic and basic sites ($139.01 \text{ kJ mol}^{-1}$ and $47.63 \text{ kJ mol}^{-1}$, respectively) with higher intrinsic activation energies ($141.05 \text{ kJ mol}^{-1}$ and $95.35 \text{ kJ mol}^{-1}$, respectively). For these proposed ES, the lower intrinsic activation energy to form the enolate intermediate suggests that this route is facile compared to the more energy-intensive deprotonation at the carbonyl group.

The deprotonation at the α -carbon produces the enolate intermediate that can later react with other adsorbed acetaldehyde molecules to form the C–C coupling. As indicated by the bond dissociation energies, the deprotonation to form enolate requires more energy than the deprotonation at the carbonyl group ($408.52 \text{ kJ mol}^{-1}$ vs $380.41 \text{ kJ mol}^{-1}$, respectively). In addition, the co-adsorption energy of the enolate is also more negative compared to the intermediate obtained from the deprotonation of the carbonyl group ($-452.81 \text{ kJ mol}^{-1}$ vs $-311.81 \text{ kJ mol}^{-1}$, respectively). The reaction pathway is depicted in Fig. 5 for both acetaldehyde deprotonation routes.

3.3.3. Aldol condensation

The formation of the C–C bond on HAP catalysts, as experimentally suggested [9,12,20], results from the nucleophilic attack of the enolate onto the carbon center of the carbonyl group in a neighboring acetaldehyde molecule. This mechanism was adopted here to model the aldol condensation on the HAP(0001) surface, Fig. 6.

In the first condensation route investigated, the enolate and acetaldehyde are adsorbed on the same Lewis acidic Ca^{2+} ion (ES 11a), resulting in the $\text{C}_4\text{H}_7\text{O}_2$ intermediate adsorbed via both of its oxygen atoms (bidentate). The C–C coupling via this route was calculated to be endothermic at $51.79 \text{ kJ mol}^{-1}$ with an intrinsic activation energy of $53.47 \text{ kJ mol}^{-1}$. Despite the lower stability

of the enolate intermediate, the co-adsorption energies of the enolate and acetaldehyde were similar ($-111.04 \text{ kJ mol}^{-1}$ vs $-109.57 \text{ kJ mol}^{-1}$, respectively) due to lateral interactions between the adsorbates.

The second condensation route involved the enolate adsorbed on the acidic site with the acetaldehyde molecule adsorbed on a neighboring basic site (ES 11b). In this configuration, the reaction has a lower intrinsic activation energy of $15.47 \text{ kJ mol}^{-1}$ and it is slightly exothermic at $-5.65 \text{ kJ mol}^{-1}$. Furthermore, as the acetaldehyde moved to a basic site, the difference between the enolate and acetaldehyde co-adsorption energies increased. The co-adsorption of the enolate energy increased to $-142.66 \text{ kJ mol}^{-1}$, while the co-adsorption energy of the acetaldehyde decreased to $-49.19 \text{ kJ mol}^{-1}$.

Finally, the $\text{C}_4\text{H}_7\text{O}_2$ intermediate from the C–C coupling reaction was protonated to form the aldol $\text{C}_4\text{H}_8\text{O}_2$ (ES 12). This reaction was calculated to be exothermic ($-85.58 \text{ kJ mol}^{-1}$) with an intrinsic activation energy of $66.14 \text{ kJ mol}^{-1}$. The protonation of the oxygen bound to the secondary carbon, Fig. 6, helps to stabilize the C–C bond formed in the aldol condensation, as the C–C bond dissociation energy increases from 6.43 kJ mol^{-1} to $321.26 \text{ kJ mol}^{-1}$. The adsorption energies on the HAP(0001) reflect this stabilization: the adsorption energy of the aldol was found to be $23.47 \text{ kJ mol}^{-1}$ greater than that for the $\text{C}_4\text{H}_7\text{O}_2$ intermediate.

3.3.4. Insights from the DFT-based results

The DFT activation energies indicate that the most energetically favorable sequence of elementary steps on the HAP(0001) surface starts with the adsorption of ethanol on basic sites. The adsorbed reactant then goes through the deprotonation of the hydroxy group to form ethoxide (ES 1b), and subsequent deprotonation of the α -carbon to acetaldehyde (ES 5). The adsorption of acetaldehyde on basic sites also favors its deprotonation to enolate (ES 9b), which later reacts with a neighboring adsorbed molecule of acetaldehyde to form the aldol. In the aldol condensation, the most energetically favorable route involves the adsorption of enolate on the Lewis acidic Ca^{2+} ion, while the acetaldehyde is adsorbed on a neighboring basic site (ES 11b). The bidentate adsorption of the $\text{C}_4\text{H}_7\text{O}_2$ intermediate occurs via both of its oxygen atoms bound to the same acidic site. The aldol is then formed by a subsequent protonation of the oxygen bound to the secondary carbon (ES 12).

The results in Section 3.2 show that the presence of both acidic and basic sites on the HAP surface favors the production of higher alcohols via the C–C bond formation. This is consistent with the intrinsic activation energies calculated for the elementary steps in the aldol condensation (ES 11a and 11b). When both aldehyde intermediates were adsorbed on acidic sites, the activation energy was approximately 3.5 times higher than when neighboring acidic and basic sites were involved. This result also agrees with experimental studies by other authors [12,20] that used *in situ* spectroscopy to confirm the need for neighboring acidic and basic sites to accelerate the aldol condensation over HAP catalysts.

For the dehydrogenation of ethanol, it has been experimentally proposed [9] that the mechanism starts with the deprotonation of the ethanol O–H group. This reaction produces the ethoxide intermediate adsorbed on a Lewis acidic Ca^{2+} ions and a proton-like hydrogen adsorbed on neighboring basic sites. The ethoxide is then deprotonated at the α -carbon by a surface oxygen to form acetaldehyde. These findings are consistent with the intrinsic activation energies calculated here. As indicated in Fig. 4, the most energetically favorable route to form acetaldehyde occurs via the same elementary steps described in the aforementioned studies [9,12].

As no impurity phase was detected in our X-ray powder diffraction analysis, the occurrence of different facets in the same HAP

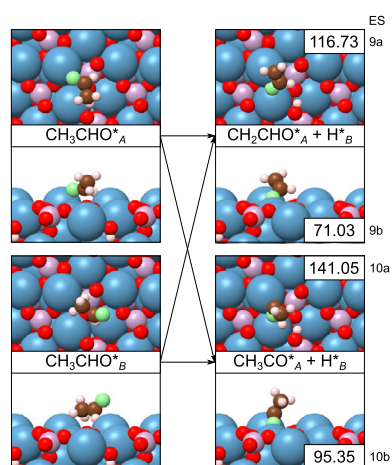


Fig. 5. Reaction pathways for acetaldehyde deprotonation on HAP(0001). Numbers inside the white rectangles represent the intrinsic activation energy (kJ mol^{-1}) for each of the ES listed in Table 6. Blue, pink, and red spheres represent Ca, P, and O atoms in the substrate, respectively. Brown, light green, and white spheres represent C, O, and H atoms in the adsorbate. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

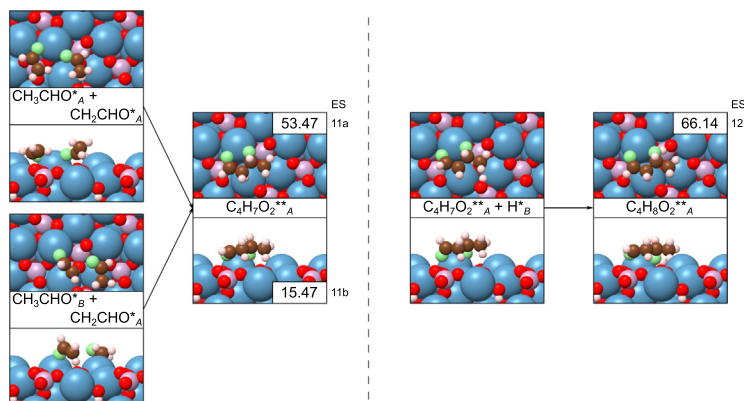


Fig. 6. Reaction pathways for aldol condensation on HAP(0001). Numbers inside the white rectangles represent the intrinsic activation energy (kJ mol^{-1}) for each of the ES listed in Table 6. Blue, pink, and red spheres represent Ca, P, and O atoms in the substrate, respectively. Brown, light green, and white spheres represent C, O, and H atoms in the adsorbate. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

crystal could explain the appearance of different Ca/P ratios on the catalyst's surface. Interestingly, experimental [74] and simulated [33] results have indicated that different HAP surface terminations have similar adsorption properties. In this regard, the choice of HAP(0001) may capture the experimental results seen in non-specific HAP surfaces. However, this observation should not be considered definitive, because the impact of non-stoichiometric and defective HAP surfaces on the conversion of ethanol needs to be addressed in future studies.

Concerning the HAP surface characterization, different species were suggested as active sites for ethanol conversion. Recently, Wang et al. [75] proposed that $\text{Ca}^{2+}/\text{OH}^-$ and POH/OH^- are the main active sites for aldol condensation and hydrogen transfer, respectively. Ho et al. [12] proposed $\text{CaO}/\text{PO}_4^{3-}$ as aldol condensation sites and Ca–O as dehydrogenation and hydrogen transfer sites. Here, although only the HAP(0001) termination was modeled as the substrate, the evaluated sequence of elementary steps is consistent with experimental results. This suggests that it is reasonable to assume Ca^{2+} ions as Lewis acidic sites and O atoms in PO_4^{3-} groups as Lewis basic sites on HAP(0001).

4. Conclusions

In this study, hydroxyapatite solids were synthesized using different post-synthesis treatments and tested for the conversion of ethanol. We found that post-synthesis treatments using autoclave, ultrasound or microwave considerably decrease the preparation time of hydroxyapatite. By using microwaves, for example, the synthesis time decreased from 24 h to 10 min. Furthermore, the use of ultrasound and microwave influences mainly the number and strength of the surface basic sites of hydroxyapatite.

Even though all hydroxyapatite solids under study had the same bulk composition, i.e., $\text{Ca/P} = 1.66$, differences in the surface Ca/P ratio led to different catalytic activity of hydroxyapatite in the ethanol conversion. In this respect, the ratio of basic to acid sites in the catalysts can be used to aid in the prediction of product selectivity. Furthermore, we found that the solids treated with microwave for 30 min have the highest turnover rates for butanol and hexanol production from ethanol and should be considered further for the preparation of hydroxyapatite-based catalysts.

We also report for the first time, to the best of our knowledge, DFT calculations for key elementary steps for the ethanol conversion on the hydroxyapatite catalyst. The substrate was modeled using the HAP(0001) termination, which could capture results seen in non-specific HAP surfaces. The calculated intrinsic activation energies are consistent with our experimental observations. The

most energetically favorable sequence of elementary steps for ethanol dehydrogenation begins with the deprotonation of the hydroxy group, followed by hydrogen abstraction of the α -carbon to form acetaldehyde. Once the acetaldehyde is formed, the deprotonation of the α -carbon leads to the formation of the enolate intermediate required for the aldol condensation. The experimental and theoretical results show that a balanced distribution of acidic and basic sites on HAP catalysts is critical to accelerate the formation of C–C bonds via aldol condensation.

5. Data availability statement

The manuscript includes all results obtained in this study and required to support our discussion and conclusions. Furthermore, complementary experimental and theoretical results (figures, tables) are available in the supplementary material. Additional data can be obtained under request directly from the authors.

Author contributions

Henrique Brasil: writing - original draft, methodology, formal analysis, investigation. **Albert F.B. Bittencourt:** writing - original draft, methodology, formal analysis, investigation, figures. **Kathleen C.E.S. Yokoo:** formal analysis, investigation, writing - review & editing. **Paulo C. D. Mendes:** writing - review & editing. **Lucas G. Verga:** writing - review & editing. **Karla F. Andriani:** writing - review & editing. **Richard Landers:** resources. **Juarez L.F. Da Silva:** supervision, writing - review & editing, project administration (theoretical). **Gustavo P. Valença:** supervision, project administration (experimental).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.jcat.2021.08.050>.

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